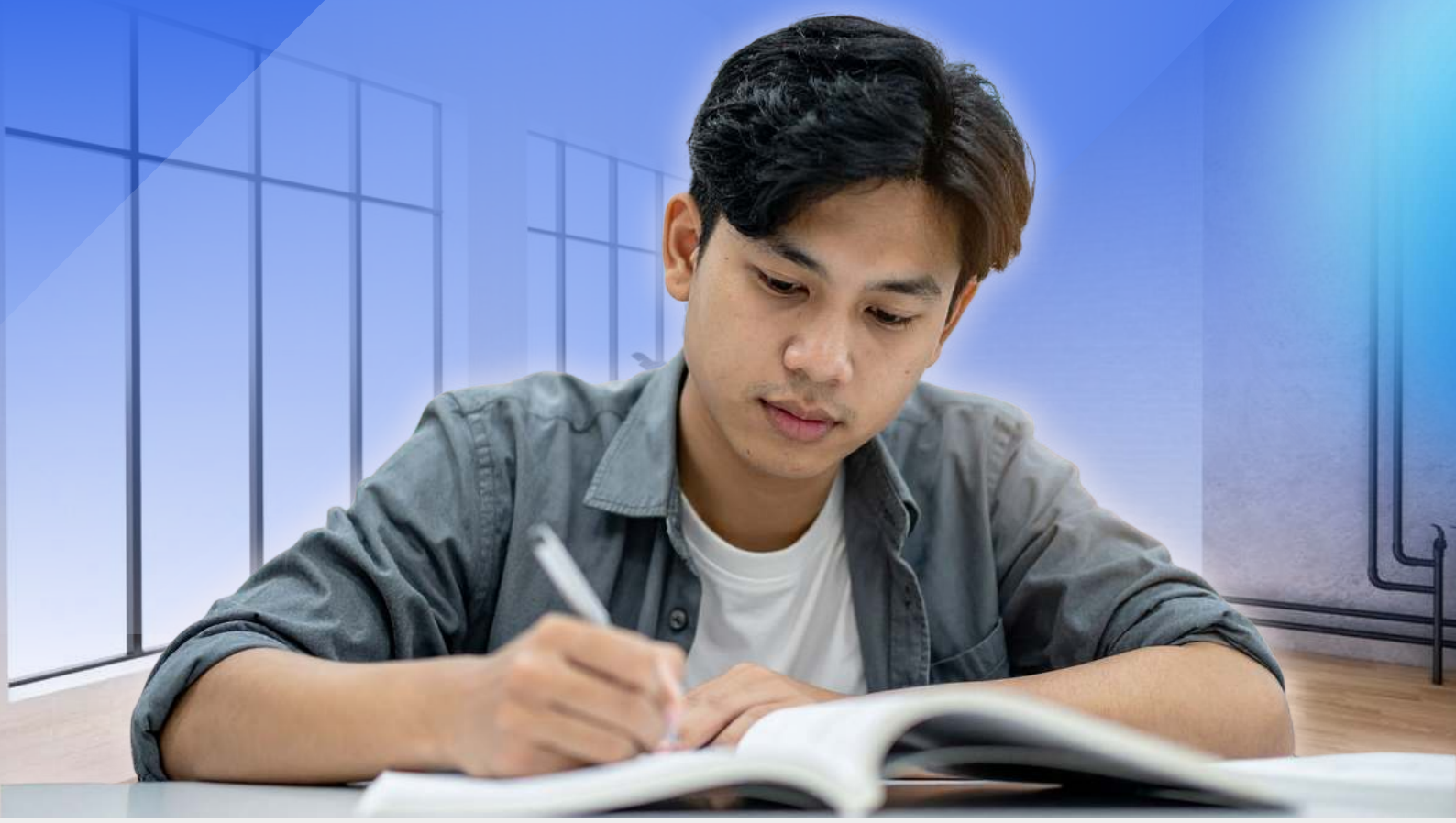
Lenterago

PRE TEST

SAT MATH

61001

44 QUESTIONS



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MODUL 1

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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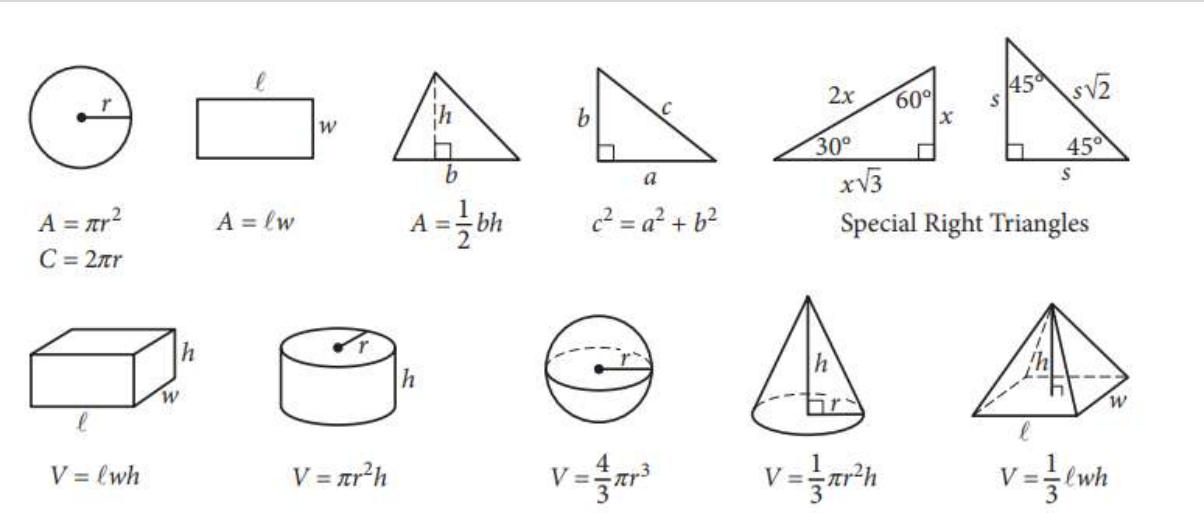
REFERENCE



The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

0 8 2 1 7 0 2 7 6 0 7 6

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Module 1

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1. Which expression is equivalent to

$$x^2 + 7x - 30?$$

A) $(x-3)(x+10)$
B) $(x-6)(x+5)$
C) $(x-10)(x+3)$
D) $(x-2)(x+15)$

2. Maria bikes at a speed of 12 miles per hour and jogs at a speed of 6 miles per hour. She bikes for b hours and jogs for $j$hours for a combined total of 48 miles.
Which equation represents this situation?

A) $12b + 6j = 48$

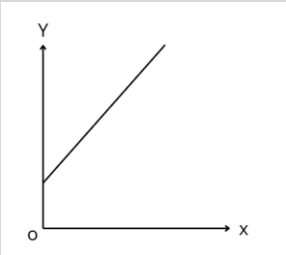
B) $\frac{1}{12}b + \frac{1}{6}j = 48$

C) $\frac{1}{12}b + \frac{1}{6}j = 288$

D) $12b + 6j = 288$

3. In triangle XYZ, the measure of angle X is 40° and the measure of angle Y is 85°.
What is the measure of angle Z?

A) 45°
B) 55°
C) 65°
D) 95°

4. 

The graph shows the total cost y, in dollars, of renting a bicycle for x hours. The rental shop charges a one-time flat fee plus an hourly rate.

What is the best interpretation of the slope of the graph?

A) The hourly rental rate for the bicycle
B) The one-time flat fee for the rental
C) The maximum total cost of the rental
D) The total cost of renting the bicycle

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5. The table summarizes the distribution of color and shape for 120 tiles of equal area.

	Red	Blue	Green	Total
Circle	15	25	20	60
Triangle	25	15	20	60
Total	40	40	40	120

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

6. From a population of 40,000 people, 800 were chosen at random and surveyed about a proposed new law. Based on the survey, it is estimated that 45% of people in the population support the law, with an associated margin of error of ±4%.

Based on these results, which of the following is a plausible value for the total number of people in the population who support the law?

A) 1,800
B) 22,000
C) 18,400
D) 35,200

7. $$\frac{60}{x+4} = x$$

What is the positive solution to the given equation?

8. A hot-air balloon descends from an altitude of 3,200 feet to 1,400 feet at a constant rate of 90 feet per minute.

What type of function best models the relationship between the balloon's altitude and time?

A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

9. $$h(x) = 20\left(\frac{1}{5}\right)^x$$

If the given function $h$is graphed in the xy-plane, where $y = h(x)$,what is the y-intercept of the graph?

A) $(0, 1)$
B) $(0, 5)$
C) $(0, 20)$
D) $(0, 25)$

0 8 2 1 7 0 2 7 6 0 7 6

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10. A circle has center O, circumference 160π, and two diameters $AC$and BD that intersect at O. Because they are diameters, opposite central angles are congruent, so the arcs they subtend in opposite positions are congruent.

    Suppose the length of arc $AD$is three times the length of arc AB. What is the length of arc BC?

    A) 20π
    B) 40π
    C) 60π
    D) 80π

11. A rectangle has a length of x units and a width of $(x-12)$ units. If the rectangle has an area of 140square units, what is the value of x?

    A) 7
    B) 14
    C) 20
    D) 28

12. Liam opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Liam opened the account and the total amount n, in dollars, in the account.

Time (years)	Total amount (dollars)
0	800.00
1	804.00
2	808.02

    If Liam made no additional deposits or withdrawals, which of the following equations best represents the relationship between $t$and n?

    A) $n = (1+800)^t$
    B) $n = (1+0.005)^t$
    C) $n = 800(1+0.005)^t$
    D) $n = 0.005(1+800)^t$

13. At how many points do the graphs of the equations

$$y = 2x + 10 \; and \; y = -3x$$

    intersect in the xy-plane?

    A) 0
    B) 1
    C) 2
    D) Infinitely many

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14. At a carnival, students can win colored tokens that are worth a different number of points depending on the color. One student won Bblue tokens and Yyellow tokens worth a total of 360 points. The given equation represents this situation:

$$6B + 36Y = 360$$

How many more points is a yellow token worth than a blue token?

15. The number of bacteria in a liquid medium triples every day. There are 10,000 bacteria in the liquid medium at the start of an observation. Which equation best represents the number of bacteria, y, in the liquid medium $t$days after the start of the observation?

A) $y = \frac{1}{3}(10{,}000)^t$
B) $y = 3(10{,}000)^t$
C) $y = 10{,}000\left(\frac{1}{3}\right)^t$
D) $y = 10{,}000(3)^t$

16. A cylinder has a diameter of 10 inches and a height of 15 inches. What is the volume, in cubic inches, of the cylinder?

A) 25π
B) 75π
C) 375π
D) $1{,}125\pi$

17. x+3y=18
$x + y = 6$

The solution to the given system of equations is (x, y) .What is the value of y?

A) 2
B) 4
C) 6
D) 8

18. In triangle JKL, angle J is the right angle. It is given that

$$\cos(K) = \frac{24}{25}.$$

What is the value of *cos(L)* ?

A) $\frac{7}{25}$
B) $\frac{7}{24}$
C) $\frac{24}{25}$
D) $\frac{7}{25}$

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19. The function is defined as

$$f(x) = 3x^2 - 24x + 40$$

For what value of $x$does $f(x)$reach its minimum?

20. In the xy-plane, line ℓpasses through the point $(0, 0)$and is parallel to the line represented by the equation

$$y = 5x + 7.$$

If line ℓalso passes through the point $(4, d)$,what is the value of d?

21. In the xy-plane, a line with equation

$$2y = c$$

for some constant $c$intersects a parabola at exactly one point. If the parabola has equation

$$y = -x^2 + 6x,$$

what is the value of c?

22. Solve for x:

$$x + 35 = 82$$

What value of $x$satisfies the equation?

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MODUL 2

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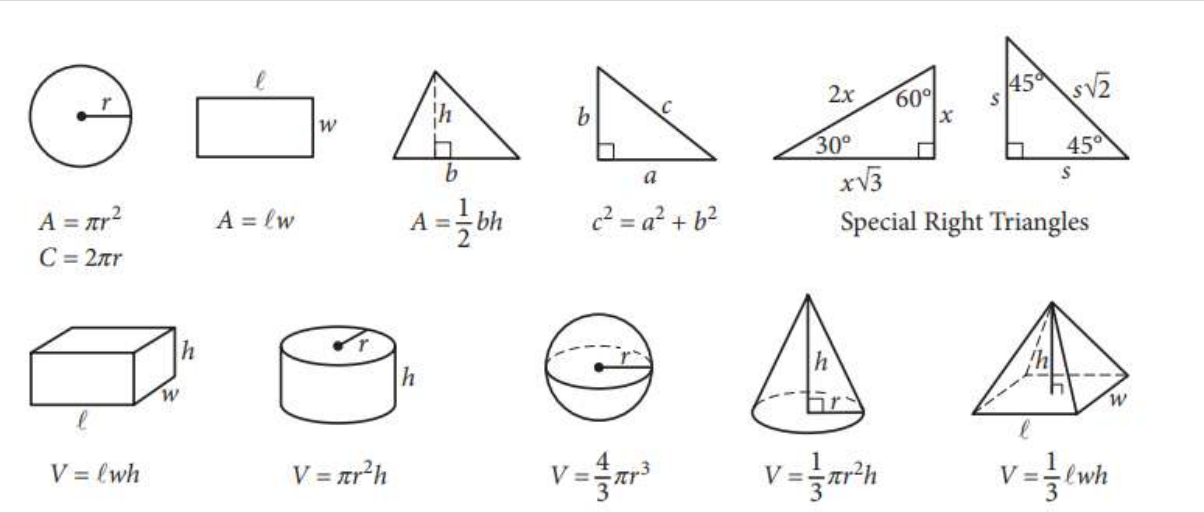

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23. The data set is:

$$60, 62, 65, 68, 72, 75, 80, 85, 90$$

What is the median of the data shown?

A) 65
B) 68
C) 72
D) 75

24. What is the area of a rectangle with a length of 15 cm and a width of 8 cm?

A) 23 cm^2
B) 60 cm^2
C) 120 cm^2
D) 240 cm^2

25. Which expression is equivalent to

$$25w - (6w + 5w)?$$

A) $14w$
B) $19w$
C) $20w$
D) $21w$

26. The number $y$is 56 less than the number x. Which equation represents the relationship between $x$and y?

A) $y = x + 56$
B) $y = x - 56$
C) $y = 56x$
D) $y = \frac{1}{56}x$

27. The expression

$$\frac{35}{7x + 56}$$

is equivalent to

$$\frac{5}{x + b},$$

where $b$is a constant and $x > 0$. What is the value of b?

A) 7
B) 8
C) 10
D) 56

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28. Out of 400 seeds that were planted, 65% sprouted.
    How many of these seeds sprouted?

29. The function f is defined by

$$f(x) = x^2 + 7.$$

What is the value of $f(4)$?

A) 15
B) 23
C) 24
D) 31

30. If $8 + x = 15,$

what is the value of $20 + 4x$?

31. The total cost $f(x)$,in dollars, to lease a car for 24 months is given by

$$f(x) = 24x + 800,$$

where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is $350?

A) $8,200
B) $8,400
C) $8,800
D) $9,200

32. The function g is defined by

$$g(x) = 12x - 5.$$

What is the value of $g(7)$?

A) 79
B) 80
C) 81
D) 82

33. Sam set a goal to run at least 30 kilometers every day to prepare for a marathon. On a certain day, Sam runs at an average speed of 5 kilometers per hour.

What is the minimum number of hours Sam must run on that day to fulfill the daily goal?

A) 4
B) 5
C) 6
D) 7

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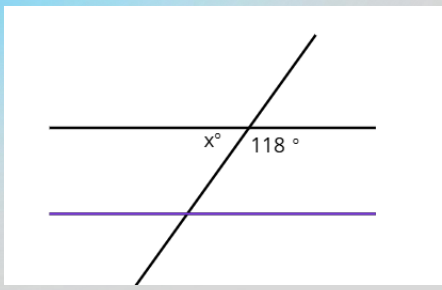
34. 

Disclaimer: Figure not Drawn to Scale

In the figure, line $j$is parallel to line k. If one angle measures 118° ,what is the value of the adjacent angle x?

35. The graph of

$$5x + 3y = -30$$

in the xy-plane has an x-intercept at $(a, 0)$and a y-intercept at $(0, b)$,where $a$and $b$are constants.
What is the value of $\frac{b}{a}$?

A) $-\frac{3}{5}$

B) $-\frac{5}{3}$

C) $\frac{3}{5}$

D) $\frac{5}{3}$

36. An object travels at a constant speed of 15 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 225 centimeters?

A) 10
B) 12
C) 15
D) 20

37. Lisa paid a total of \$205 for a tablet by making a down payment of \$25 plus $p$monthly payments of \$20 each.
Which of the following equations represents this situation?

A) $20p + 25 = 205$
B) $25p + 20 = 205$
C) $20p - 25 = 205$
D) $25p - 20 = 205$

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38. An object is thrown upward from a platform. The equation

$$h = -5t^2 + 8t + 20$$

represents this situation, where $h$is the height of the object above the ground (in meters), and $t$is the time (in seconds) after it was thrown.

Which number represents the height, in meters, from which the object was thrown?

A) 8
B) 20
C) 28
D) 40

39. The table shows three values of $x$and their corresponding values of y. There is a linear relationship between $x$and y.

When $x = 0$, $y = 12$

When $x = 1$, $y = 9$

When $x = 2$, $y = 6$

Which of the following equations represents this relationship?

A) $y = 3x + 12$
B) $y = -3x + 12$
C) $y = 6x + 12$
D) $y = -6x + 12$

40. The function is defined as

$$h(x) = x^2 - 2$$

Which table gives three values of $x$and their corresponding values of $h(x)$for the given function?

A)

x	$h(x)$
1	-1
2	2
3	7

B)

x	$h(x)$
1	-2
2	2
3	7

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C)

x	$h(x)$
1	1
2	4
3	9

D)

x	$h(x)$
1	-1
2	0
3	2

41. In the linear function f, $f(0) = 5$and $f(2) = 13$ .Which equation defines f?

A) $f(x) = 4x + 5$
B) $f(x) = 5x + 5$
C) $f(x) = 8x + 5$
D) $f(x) = 13x$

42. The given equation relates the numbers j, k,and m:

$$10j + 4k = m$$

Which equation correctly expresses $k$in terms of $j$and m?

A) $k = \frac{m-10j}{4}$
B) $k = \frac{10j-m}{4}$
C) $k = \frac{m}{10j+4}$
D) $k = 10j + m$

43. The side lengths of right triangle $XYZ$are given:

$$XY = 21,\ YZ = 28,\ XZ = 35$$

Triangle $XYZ$is similar to triangle PQR, where Ycorresponds to $Q$and Zcorresponds to R. What is the value of tan(R)?

A) $\frac{21}{28}$
B) $\frac{28}{21}$
C) $\frac{21}{35}$
D) $\frac{28}{35}$

0 8 2 1 7 0 2 7 6 0 7 6

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44. Data:

5,5,5,7,7,9,9,9,9

Which frequency table correctly represents the data listed?

A)

Number	Frequency
5	3
7	2
9	4

B)

Number	Frequency
5	2
7	3
9	4

C)

Number	Frequency
5	3
7	4
9	2

D)

Number	Frequency
5	9
7	7
9	5

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